

Fine-Tuning Large Language Models for Character Consistency Using Synthetic Datasets

Abstract

This study investigates whether a smaller language model, adapted through parameter-efficient fine-tuning, can reproduce the emotionally coherent and personality-consistent behavior of a fictional character—in this case, Harry Potter—with fidelity comparable to a large foundation model such as GPT-4o. To this end, a synthetic dialogue dataset was generated using the GPT-4o API at a cost of approximately 10 €, capturing the emotional tone, interpersonal attitudes, and contextual nuances of the character as depicted in *Harry Potter and the Philosopher’s Stone*. This dataset serves as a distilled representation of GPT-4o’s expressive and interpersonal knowledge, forming a compact training corpus for persona conditioning. A smaller model (LLaMa 3.2 3B/7B) was subsequently fine-tuned using Low-Rank Adaptation (LoRA) to internalize and consistently express the target character’s emotional and relational traits across diverse conversational contexts.

Evaluation was conducted through a controlled comparison in which GPT-4o and the fine-tuned model received identical system prompts and event-specific contexts. Their outputs were assessed for coherence, emotional realism, faithfulness to character, and naturalness of behavior. Results indicate that the fine-tuned model preserves stylistic and emotional continuity similar to GPT-4o while requiring a fraction of the computational resources, with training remaining feasible even on consumer GPUs when context lengths are constrained.

These findings suggest that targeted persona fine-tuning with synthetic data can effectively distill expressive and contextual capabilities from large models into smaller architectures, enabling lightweight systems to exhibit stable, believable, and human-like character behavior.

The project’s code and synthetic dataset can be accessed at <https://github.com/Dominik-CH/hs2025>.

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1 Introduction

Large language models (LLMs) have rapidly transformed the way humans interact with digital systems. From customer support and

educational tutoring to therapeutic dialogue, role-playing applications, and creative storytelling, conversational agents are increasingly integrated into everyday life. As these systems become more prevalent, expectations regarding their authenticity, emotional intelligence, and personality consistency continue to grow. [1] Users no longer engage with chatbots as static tools but as interactive entities that should exhibit stable behavior, coherent emotional tone, and recognizable personality traits across interactions.

However, current large-scale models such as GPT-5, while capable of producing fluent and contextually rich dialogue, often struggle to maintain a consistent persona over time. Their tone, emotional warmth, and style of communication can fluctuate between sessions, leading to interactions that feel inconsistent or impersonal. A striking real-world example of this phenomenon occurred when OpenAI released GPT-5: many users reported that the new model appeared “colder,” “less empathetic,” and “more detached” compared to GPT-4 [2]. The community response was strong enough that OpenAI subsequently reintroduced GPT-4 as an available model option, highlighting how sensitive users are to perceived changes in personality and emotional tone. This case illustrates the critical role of consistency and warmth in user satisfaction and underscores that users attribute personality to AI systems—even when such traits emerge unintentionally. [3]

Such instability undermines user trust and disrupts the illusion of conversing with a believable character or companion. In domains such as storytelling, therapeutic support, or long-term collaboration between humans and AI, inconsistency in personality or emotional tone weakens authenticity and reduces engagement. As conversational AI systems increasingly act as social, narrative, and assistive partners, maintaining a coherent and believable personality becomes essential for fostering trust, empathy, and sustained interaction.

At the same time, the computational and environmental costs of deploying ever-larger models continue to rise, as Dauner et al. show a strong correlation between model scale, token generation, and CO₂ emissions [4]. This growing demand motivates the exploration of smaller, more efficient models that can be fine-tuned for specific use cases without sacrificing quality. By distilling the expressive and contextual knowledge of large models into compact architectures through techniques such as Low-Rank Adaptation (LoRA) and synthetic data generation, researchers can create specialized systems that are both sustainable and highly adaptive.

Despite rapid progress in large-scale model capabilities, little research has systematically examined whether persona-specific synthetic datasets can be used to transfer the emotional coherence and relational depth of large models into smaller architectures. Existing work largely focuses on general fine-tuning performance or style control, but rarely investigates the stability of character-based emotional expression and consistent personality reproduction across diverse contexts.

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This work investigates whether persona-specific synthetic dialogue data, combined with Low-Rank Adaptation, can induce stable and emotionally coherent behavior in a compact language model.

The motivation behind this study is to examine the practical feasibility of transferring the emotional and interpersonal traits of a well-defined fictional character into a smaller language model. Using a synthetic dataset generated via the GPT-4o API, the research explores the extent to which persona fidelity and emotional stability can be achieved through targeted fine-tuning. The ultimate goal is to demonstrate that smaller, environmentally conscious models can exhibit stable, human-like character behavior comparable to that of larger systems, thereby contributing to both sustainable AI development and more authentic human-agent interaction.

This paper contributes an experimental framework for generating persona-specific synthetic data and fine-tuning smaller models using LoRA, enabling a systematic evaluation of emotional consistency, personality coherence, and contextual adaptability relative to large-scale baselines.

2 Background

2.1 Large Language Models and Transformers

Large language models (LLMs) such as GPT-4o are built upon the Transformer architecture [5], which has become the dominant framework for modern natural language processing. Transformers consist of stacked layers of multi-head self-attention and feed-forward networks, enabling them to encode rich contextual dependencies across sequences of arbitrary length.

At their core, Transformers rely on the *self-attention mechanism*, a process that computes for each token a context-aware representation by weighting its interactions with all other tokens in the sequence. Each token is projected into a *query* (Q), *key* (K), and *value* (V) vector. Attention scores are obtained through a scaled dot product between queries and keys, and the resulting weights determine how strongly each value vector contributes to the final representation. This allows the model to dynamically focus on semantically or syntactically relevant parts of the input.

Because self-attention operates in parallel across the entire sequence, Transformers can capture global dependencies, cross-sentence relations, and subtle linguistic cues with high efficiency. These properties make them exceptionally well-suited for language modeling: during pretraining on large-scale text corpora, Transformers learn deep statistical patterns spanning semantics, syntax, discourse structure, and long-range contextual relationships.

Transformers excel particularly in tasks requiring:

- **context integration** across long or complex sequences,
- **flexible conditioning** on arbitrary spans of text,
- **modeling of nuanced semantic and stylistic patterns**,
- **generation of coherent multi-step reasoning or dialogue**.

These strengths are central for this work, as persona modeling requires the ability to track stylistic signals, emotional tone, and character-specific behavior across multi-turn interactions—capabilities that emerge naturally from the Transformer’s attention-driven architecture. For comprehensive technical introductions, see [5].

2.2 LoRA: Low-Rank Adaptation

Large language models contain billions of parameters, making full fine-tuning computationally expensive and vulnerable to overwriting pretrained knowledge. [6] Low-Rank Adaptation (LoRA) [6] addresses these limitations by introducing a lightweight mechanism that learns small, task-specific updates while keeping the original model weights fixed. Because only a compact set of new parameters is optimized, LoRA enables rapid specialization on consumer hardware and supports a modular design in which multiple adapters—such as different personas or domain-specific behaviors—can be stored, exchanged, and activated without retraining the base model [6]. This makes LoRA particularly attractive for persona-centered applications, where stylistic or behavioral characteristics must be added without altering the model’s general linguistic abilities.

2.2.1 Concept and Mechanism. LoRA decomposes the update to a pretrained weight matrix into a low-rank representation. Instead of modifying the full weight W_0 , the model learns a low-dimensional update $\Delta W = BA$, where the trainable matrices A and B contain far fewer parameters than the original layer. The effective weight used during inference is therefore $W = W_0 + \Delta W$ (see Fig. 1). This approach allows the model to acquire new behavioral patterns while preserving the linguistic and semantic knowledge encoded in the frozen backbone.

In Transformer models, LoRA is typically applied to the attention projection matrices—especially the query and value projections that determine how the model allocates and integrates attention. Adjusting these components enables fine-grained modulation of stylistic, emotional, or persona-specific tendencies while leaving the broader semantic and syntactic competence of the model unaffected.

2.2.2 What Gets Adapted and How It Affects the Base Model. By inserting low-rank adapters into selected attention layers, LoRA reshapes the internal attention dynamics that govern how contextual cues are interpreted. The pretrained backbone remains unchanged, meaning that the model’s general linguistic and world knowledge are preserved. Persona- or task-specific behavior is instead layered on top of these capabilities: the model expresses the desired stylistic or emotional characteristics when an adapter is active, but retains full general-purpose behavior when the adapter is removed. This ensures that LoRA acts as a targeted specialization rather than a modification of the underlying model identity.

2.2.3 The Role of Rank. The rank r controls the expressive capacity of the LoRA update. Smaller ranks constrain the number of learnable parameters, which stabilizes training and reduces the computational footprint but may limit the model’s ability to represent subtle stylistic or emotional patterns. Higher ranks, in contrast, allow more flexible updates but increase memory usage and can lead to overfitting—particularly when training on limited or highly structured datasets. Choosing an appropriate rank therefore requires balancing expressiveness with robustness and ensuring that the model capacity matches the complexity of the target behavior. [7]

2.2.4 Why LoRA Requires Less Data. Since LoRA modifies only a narrow set of task-specific parameters, it relies on the pretrained

backbone for virtually all aspects of general language understanding. Fine-tuning therefore focuses solely on learning the behavioral deviations from the base model—such as persona-consistent tone, emotional tendencies, or stylistic conventions—rather than relearning linguistic structure from scratch. This targeted adaptation substantially reduces data requirements [6] and makes LoRA effective even when only small, high-quality synthetic datasets are available.

2.2.5 Advantages over Full Fine-Tuning. LoRA offers many of the benefits of full fine-tuning while requiring only a fraction of the trainable parameters, as demonstrated by Hu et al. [6] and confirmed in subsequent comparative studies [7]. By leaving the pre-trained weights untouched, it avoids catastrophic forgetting and maintains the model's general reasoning and linguistic abilities. The reduced computational cost enables faster iteration, and the modular structure of LoRA adapters allows different personas or task-specific behaviors to coexist without retraining the underlying model. This makes LoRA particularly suited for applications that require efficient and repeatable specialization, such as producing distinct character or domain-specific variants of a base model.

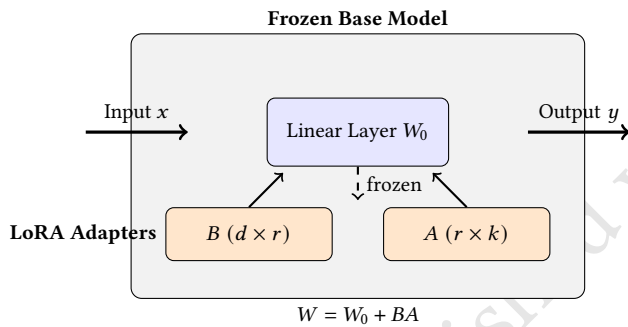


Figure 1: LoRA augments a frozen weight matrix W_0 with a trainable low-rank update $\Delta W = BA$, where B projects from d to r dimensions and A projects from r to k dimensions. Only A and B are optimized during fine-tuning.

2.3 Synthetic Data and Persona Fine-Tuning

Synthetic data refers to artificially generated datasets created by larger models to simulate real, domain specific data. In case of conversational data, instead of collecting and labeling human dialogue, researchers can use advanced LLMs to produce controlled outputs that capture specific linguistic or behavioral patterns. By adjusting prompt design, sampling temperature, and output constraints, it is possible to generate highly consistent and diverse datasets tailored to a given domain or in this case persona.

This approach is effective for fine-tuning smaller or distilled models. The synthetic data functions as a distilled representation of the larger model's expressive and contextual capabilities, transferring its learned patterns into a more compact form. When combined with fine-tuning methods such as LoRA, the smaller model can adopt the emotional tone, interpersonal style, and situational adaptability encoded in the synthetic data. As a result, the fine-tuned model gains the ability to produce responses that remain coherent,

context-aware, and stylistically consistent, despite having far fewer parameters than its source model.

2.4 Personality Consistency in Human-Agent Interaction

While large language models exhibit strong linguistic fluency, current dialogue systems still struggle to maintain continuity in their emotional understanding across interactions. As Debnath argues, truly natural and empathetic agents must retain and use information about a user's prior emotional states, yet many models lack mechanisms for sustaining such affective memory over time [8].

Findings in human-agent interaction research further show that personality cues can shape how users engage with artificial agents. For example, Krocze et al. demonstrate that agents displaying more extroverted behavioral traits are more likely to be approached for assistance in collaborative tasks [9]. This highlights the functional importance of stable, recognizable personality expressions.

Achieving personality-consistent behavior in LLMs is therefore not only a technical aspiration but a prerequisite for creating digital characters that users treat as reliable and socially meaningful counterparts.

2.5 Connecting Technical Methods to Persona Modeling

This work combines LoRA-based fine-tuning with synthetic data generation to explore whether a smaller model can reproduce the distinctive personality and emotional tone of a specific fictional character of Harry Potter. The approach aims to transfer the character's interpersonal attitudes, and context-dependent emotional reactions into a compact, efficient model. By aligning low-rank adaptation with persona-specific data, the study bridges technical efficiency and affective modeling, contributing to research on emotionally coherent, personality-consistent language agents. Beyond its technical implications, this work advances understanding of how fine-tuned LLMs can embody stable, human-like identities within interactive settings.

3 Related Work

Research on character-centric dialogue systems has expanded substantially with the increasing capabilities of large language models. Prior and subjectively selected work based on the PRISMA systematic literature review relevant for this project may be organized into three domains.

3.1 Integrated Character Systems

A body of work focuses on developing end-to-end systems that support coherent, personality-driven interactive agents. [10] introduces a comprehensive architecture unifying conversational modeling, emotional and personality management, memory mechanisms, voice synthesis, animation, and environment interaction to enable believable digital characters.[8] The "Digital Einstein" demonstration highlights the challenges inherent in coordinating diverse AI subsystems to maintain a consistent character identity. Similarly, [11] proposes a full-process pipeline combining semi-automated

character data construction, parameter-efficient fine-tuning, explicit knowledge-boundary control, and a dual-term memory mechanism. The authors report improved persona adherence and reduced hallucinations, indicating the critical role of structured background knowledge and memory in character-centric dialogue generation. These contributions collectively underscore that reliable character agents require more than language modeling alone, benefiting from architectures that integrate memory, knowledge constraints, and purposeful fine-tuning.

3.2 Datasets and Benchmarks for Character Alignment

Another major research direction involves constructing datasets and benchmarks that facilitate the training and evaluation of character-aligned dialogue systems. *Large Language Models Meet Harry Potter* [12] presents the HPD dataset, a bilingual and richly annotated corpus capturing dialogue scenes, speaker identities, character relationships, and attributes, **that will also be used for training in regards to this project**. Experimental findings reveal that even state-of-the-art LLMs retain considerable limitations in maintaining character faithfulness across extended dialogues. Further advancing this area, *RoleLLM* introduces RoleBench, a large-scale benchmark comprising 168,093 character-aligned samples generated through role profiling, contextual knowledge extraction, and role-conditioned instruction generation [13]. Efforts to enhance long-term conversational engagement are exemplified by *KEEP CHATTING!* [14], which constructs the CONTINUOUSCHAT dataset through persona expansion, emotion-rich dialogue generation, and style-controlled rewriting to promote sustained user interaction. In the domain of empathetic interactions, *SweetieChat* develops the ServeForEmo dataset and a strategy-enhanced role-playing framework designed to generate more nuanced emotional support and scenario diversity[15]. Collectively, these datasets provide structured resources essential for evaluating both the stylistic fidelity and behavioral coherence of character-driven language models.

3.3 Methods for Persona Consistency and Stylistic Control

A third line of research proposes methods for enhancing persona consistency, improving stylistic control, and enabling computationally efficient adaptation of large models. [16] applies Direct Preference Optimization (DPO) by constructing pseudo-negative samples through persona swapping. This preference-based approach yields noticeable improvements in persona stability and dialogue naturalness compared to supervised and entailment-guided baselines. *Enhancing Persona Consistency with Large Language Models* presents a pipeline for extracting structured character descriptors, such as relationships, contextual cues, and storylines and applying Low-Rank Adaptation (LoRA) for cost-effective fine-tuning on generated dialogue corpora. The resulting models exhibit enhanced self-awareness, robustness, and individuality[17]. The comparative study *Exploring Few-Shot Learning: Fine-Tuning vs. In-Context Learning and Parameter-Efficient Adaptations* evaluates full fine-tuning, in-context learning, and LoRA across multiple tasks, concluding that LoRA provides a favorable trade-off between computational

demands and model performance, particularly in low-resource settings[18].

Overall, existing literature demonstrates notable progress in dataset construction, style and persona modeling, parameter-efficient fine-tuning, and system-level integration. These works provide important methodological insights and technical foundations for constructing character-aligned dialogue agents and inform the approach taken in this study.

4 Methods

This section presents the methodological framework for fine-tuning a compact language model to reproduce the emotional tone, conversational style, and personality traits of the fictional character *Harry Potter*. The central objective is to assess whether a smaller, LoRA-adapted model can maintain persona-consistent behavior comparable to a large-scale model such as GPT-4o.

The project is built on a synthetically generated dialogue dataset. To construct this dataset, *Harry Potter and the Philosopher's Stone* is first segmented into individual chapters. Each chapter is then processed through a prompt designed to extract all relevant events, contextual details, involved characters, and their implied emotional states. These extracted event summaries serve as structured context for generating simulated dialogues between Harry Potter and the user. Synthetic data generation is performed using the OpenAI API, with GPT-4o serving as the base model responsible for creating the dialogues. Fine-tuning is performed on the LLaMA 3.2 Instruct model (3B or 7B parameters)[19] using the Hugging Face transformers [20] and peft libraries. All implementation steps follow open-source best practices to ensure transparency and reproducibility. The complete project code can be accessed at <https://github.com/Dominik-CH/hs2025>

4.1 Modeling Setup

The model is implemented as a CausalLM, the standard autoregressive architecture for next-token prediction. It attends only to preceding tokens, enforcing a unidirectional dependency structure suitable for text generation tasks such as dialogue modeling. This contrasts with bidirectional encoders like BERT [21], which are optimized for comprehension rather than generation.

Causal LLMs naturally capture the sequential nature of dialogue: each new utterance is generated based on the full conversational history, allowing consistent emotional and stylistic continuity. Each training instance represents a short exchange, where *Harry Potter's* replies serve as target completions. Dialogues are formatted according to the LLaMA 3.2 input specification with explicit role and conversational turns, ensuring the model can differentiate between speakers. As an example of how the conversational data is transformed into the supported format, refer to Listing 1 and Table 1, which illustrate the complete formatted text and function of each token.

Listing 1: Dialogue formatting scheme used during training.

```
<|begin_of_text|>
<|start_header_id|>system<|end_header_id|>
Context or scene background
<|eot_id|>

<|start_header_id|>User<|end_header_id|>
User utterance
<|eot_id|>

<|start_header_id|>Harry Potter<|end_header_id|>
Character response
<|eot_id|>
<|end_of_text|>
```

Token	Placement	Description
< begin_of_text >	Start of sample	Marks beginning of a sample.
< start_header_id >	Before role	Begins a speaker or system header.
< end_header_id >	After role	Closes the header tag.
< eot_id >	After utterance	Indicates end of a turn and stops generation at inference.
< end_of_text >	End of sample	Marks the end of a sample.

Table 1: Conversation formatting tokens for LLaMA 3.2.

4.2 Implementation and Reproducibility

All experiments are executed in a Jupyter Notebook environment, which is published alongside this paper for reproducibility. The full pipeline—dataset loading, tokenization, fine-tuning, and evaluation—is implemented with the transformers and peft libraries. LoRA adapters are trained with varying rank and alpha parameters following guidelines by Raschka [22]. After training, adapter weights are merged into the base model and exported for inference.

4.3 Dataset and Length Variation

The dataset comprises multi-turn dialogues of varying length. Each sample preserves explicit speaker roles and turn order. To enhance diversity and prevent overfitting to a fixed input size, dialogues are segmented into overlapping windows of different lengths while retaining chronological continuity.

4.3.1 Conversation Windowing. Each conversation is split into overlapping windows:

- **Window size:** 20–30 messages (context-depth vs. efficiency tradeoff)
- **Overlap:** 30–50% of window size (e.g., for a 25-message window, overlap 10–12 messages)

This windowing technique ensures that each sample maintains narrative continuity while exposing the model to multiple starting points within the same conversation and therefore helps the model to learn long-time contextual dependencies when the starting context window is exceeded.

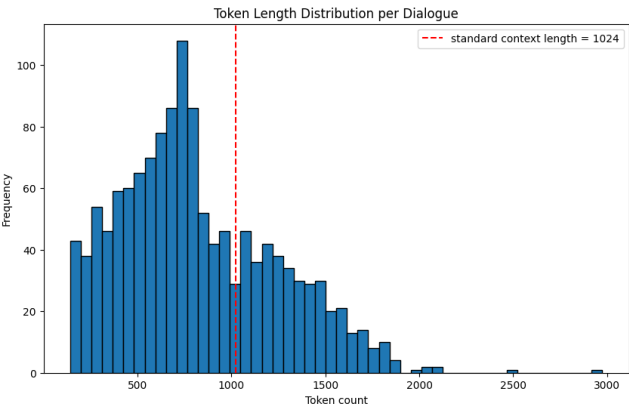


Figure 2: Histogram of total token lengths across all synthetic dialogue samples, computed using a double-tokenization procedure to accurately capture real context requirements. Each bin spans 50 tokens, and bar heights indicate the frequency of conversations within the corresponding range. This distribution was used to determine an appropriate maximum context window for fine-tuning while minimizing truncation and preserving the integrity of multi-turn interactions.

4.3.2 Mixed-Length Composition. A combination of sample lengths ensures robust generalization:

- 60–70% windowed segments (20–30 messages) which teach local continuity and dialogue flow.
- 20–30% full-length conversations which preserve long-term narrative coherence.
- 10–20% short exchanges (5–10 messages) which encourage adaptability to varied session lengths.

This mixture balances local flow, long-term narrative retention, and adaptability to brief user interactions.

4.4 Token Length Analysis

To determine an appropriate context length for fine-tuning, we first analyzed the token-length distribution of all dialogue samples. Since the dataset is composed of multi-turn interactions with variable lengths, understanding the distribution of total tokens per conversation is essential for choosing a suitable context window and avoiding truncation.

Each dialogue was tokenized twice: first, to measure token-length distributions across the dataset; and second, during preprocessing for model input generation. This double-tokenization approach, while computationally intensive, provides an accurate estimate of real context requirements and ensures no significant information loss due to sequence truncation.

A histogram of total token counts was plotted, where each bin represents a range of 50 tokens (i.e., $50 \times n$). The height of each bin indicates the frequency of conversations falling within that range. This visualization helps identify the most common token lengths and detect outliers that exceed the intended context window 3.

To complement the histogram, the Cumulative Distribution Function (CDF) was calculated. The CDF shows the cumulative fraction

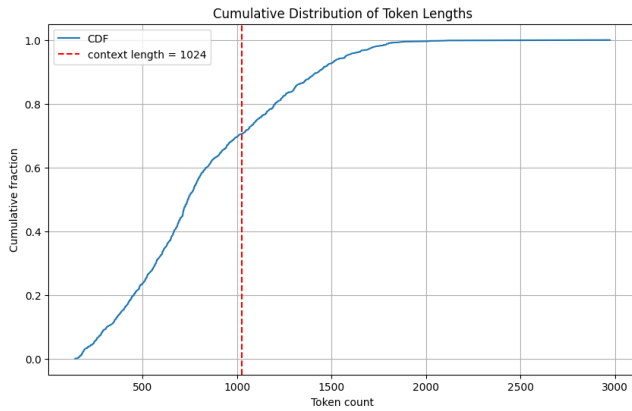


Figure 3: Cumulative distribution function (CDF) of dialogue token lengths, illustrating the proportion of conversations that fall below a given token count. This visualization complements the histogram analysis by revealing how quickly the dataset accumulates toward the upper token range and by highlighting the threshold at which only a small fraction of samples exceed a chosen context window. The CDF supports the selection of a context length that balances full coverage with computational efficiency.

of dialogues with a token count less than or equal to a given value, providing insight into how data accumulates across token ranges. As an example, for histogram bin counts [3, 7, 5, 2], the cumulative counts are [3, 10, 15, 17], and the normalized CDF values are approximately [0.176, 0.588, 0.882, 1.0]. This representation allows estimation of the proportion of samples fitting within specific token limits.

4.4.1 Findings and Context Length Decision. The analysis revealed that approximately 75% of all dialogues fall below the standard context limit of 1024 tokens. To balance coverage and efficiency, dialogues surpassing the maximum allowable sequence length - now defined to be 1250 as it covers 80% of all conversation samples - were excluded from the fine-tuning dialogue corpus. The result is a loosely Gaussian-distributed dataset that emphasizes mid-range conversational lengths while still providing adequate coverage of both shorter and longer sequences, ensuring adaptability. See Figure 4

Since the model requires uniform sequence lengths within each batch, all shorter dialogues are padded to the defined maximum using the model's designated padding token. This maintains consistent input dimensionality during training. The final preprocessing step thus guarantees that every dialogue sample fits within the optimized context window, enabling efficient and stable fine-tuning.

4.5 Training Process

The Fine-tuning process follows a standard causal language modeling (CLM) setup with parameter-efficient LoRA adapters. A total of 569 formatted dialogues were used, each containing multiple turns between *Harry Potter* and a user.

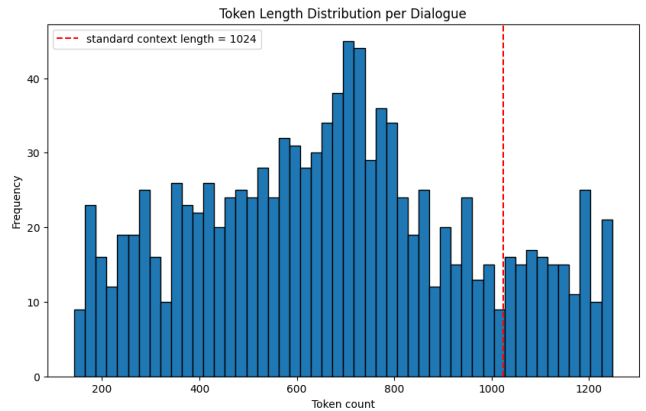


Figure 4: Histogram of token lengths after applying a maximum sequence cap of 1250 tokens. Dialogues exceeding this threshold were excluded to balance dataset coverage. The resulting distribution retains a loosely Gaussian shape, emphasizing mid-range conversation lengths while preserving sufficient variability for model adaptability in short to long conversation lengths.

4.5.1 Tokenization and Padding. Dialogues were tokenized with the LLaMA tokenizer. Dynamic padding ensures equal sequence lengths within batches, while masked padding tokens (-100) are excluded from the loss computation. Each tokenized sample consists of `input_ids`, `attention_mask`, and `labels`, maintaining context alignment and dialogue semantics.

4.5.2 Data Collation and Loader. Batch collation was handled using `DataCollatorForLanguageModeling` with `mlm=False`. The data was formatted into PyTorch tensors using `set_format("torch")` for integration with the Trainer API.

4.5.3 Model and LoRA Configuration. The base model (LLaMA 3.2 Instruct) includes 16 decoder layers, 32 attention heads, and a 2048-dimensional hidden state. LoRA adapters were added to key attention and feed-forward layers with the following configuration:

- `task_type = TaskType.CAUSAL_LM`
- `r = 8, lora_alpha = 16, lora_dropout = 0.05`
- `bias = "none"`
- `target_modules = [k_proj, q_proj, v_proj, o_proj, up_proj, down_proj, gate_proj]`

This setup adds roughly 1,280,903,168 trainable parameters (3.5% of total weights), achieving substantial savings in memory and computational resources.

4.5.4 Training Configuration. Training employed the Hugging Face Trainer with the following hyperparameters:

- `learning_rate = 2e-4` (cosine decay)
- `num_train_epochs = 100` (with early stopping)
- `per_device_train_batch_size = 4`
- `gradient_accumulation_steps = 8` (effective batch size 32)
- `warmup_ratio = 0.05, weight_decay = 0.01`
- `optimizer = "adamw_8bit"`

Training was performed on an NVIDIA H100 SXM. These hyperparameters are heavily oriented at the findings of Sebastian Raschka [22].

4.5.5 Monitoring and Early Stopping. Validation was performed once every 25 steps (`eval_strategy="steps"`), and training metrics were logged every 25 steps. A custom stopping callback was implemented to ensure stable optimization behavior: training was halted whenever the training loss failed to decrease while the validation loss simultaneously increased. This enforced the criterion that effective training requires a continuously declining training loss without concurrent degradation in validation performance. The best-performing model checkpoint was automatically restored at the end of training.

Computation of Training and Validation Loss. In a Causal Language Modeling setup, the loss represents how well the model predicts the next token in a sequence given all previous tokens. For each sequence of tokens $x = [x_1, x_2, \dots, x_T]$, the model computes a probability distribution $P(x_t | x_{<t})$ over the vocabulary for each position t . The objective is to minimize the negative log-likelihood (NLL) of the true token sequence:

$$\mathcal{L} = -\frac{1}{T} \sum_{t=1}^T \log P(x_t | x_{<t})$$

During training, this loss is averaged across all tokens in a batch and across all batches in an epoch, yielding the **training loss**. Padding tokens are ignored by assigning them the label value `-100`, ensuring that non-informative positions do not contribute to the loss.

For **validation loss**, the same computation is applied to the held-out validation set without gradient updates. The model processes validation batches in inference mode (`torch.no_grad()`), predicting next-token probabilities for each sequence. The mean validation loss thus quantifies the model's ability to generalize to unseen dialogue samples.

Both training and validation losses are reported as per-token cross-entropy values. A steadily decreasing training loss with a stable or slightly decreasing validation loss indicates effective learning, while a divergence between the two suggests overfitting and triggers the early-stopping mechanism.

4.5.6 Model Export. Finally, LoRA weights were merged with the base LLaMA parameters to form a consolidated model. The checkpoint, tokenizer, and logs were saved for later inference and evaluation against GPT-4o on measures of personality consistency, emotional tone, and contextual fidelity.

5 Analysis

The experimental results indicate that the LoRA-based fine-tuning procedure successfully transferred persona-specific linguistic and emotional patterns into the smaller base model. Across the evaluated test prompts, the fine-tuned model demonstrated recognizable traits of the target character, including stylistic phrasing, characteristic emotional tone, and consistent interpersonal attitudes derived from the synthetic dataset. While the generated outputs do not fully match the expressive depth of a larger baseline system such as GPT-4o, the smaller model - most of the time - exhibited a stable

persona and produced contextually appropriate emotional cues with convincing quality given its parameter budget.

Hypothesis. The observed lack of a substantial decrease in validation loss while training is a consequence of the persona-oriented training objective. The fine-tuning process does not aim to optimize for deterministic input-output relationships or categorical correctness, but rather to bias the model toward a characteristic style, emotional tone, and interpersonal behavior. For any given prompt, multiple semantically and stylistically valid responses exist, none of which can be considered uniquely correct. As a result, token-level validation loss constitutes a weak and potentially misleading proxy for evaluating persona alignment.

Under this hypothesis, a pronounced reduction in validation loss would primarily be expected in the presence of data leakage or excessive overlap between training and validation sets, where the model learns to react to specific surface-level phrases in fixed ways. Such behavior would indicate memorization rather than generalizable character expression and would therefore be undesirable for the intended application.

Achieving a significant reduction in validation loss under this objective would likely require an orders-of-magnitude increase in training data to cover a substantially larger space of paraphrases and semantically equivalent formulations. This requirement would fundamentally challenge the motivation for using Low-Rank Adaptation (LoRA), whose primary advantage lies in enabling effective specialization with limited, high-quality data. Full fine-tuning with large-scale domain-specific datasets would be more suitable for minimizing validation loss, but would contradict the efficiency, modularity, and sustainability goals central to this approach.

From a practical perspective, the synthetic dataset-generated entirely using the GPT-4o API through structured prompt templates—proved sufficient to enable this behavioral alignment. The generation of the complete dataset incurred a cost of approximately 10 €, making the approach highly economical compared to human-annotated alternatives. The resulting dialogue set exhibited sufficient internal consistency and stylistic variation to train a compact persona-focused model capable of predictable and repeatable character behavior.

5.1 Evaluation Protocol and Test Scenarios

Evaluation was conducted using a controlled qualitative comparison protocol. A fixed set of $N = 12$ evaluation prompts was designed based on unseen narrative situations inspired by *Harry Potter* and *the Philosopher's Stone*. Each prompt represents a short, self-contained scene requiring an emotionally grounded response rather than factual recall. Concrete inference and scoring results based on the following criteria are provided in Appendix A.

For each test case, both GPT-4o (Web) and the fine-tuned local model received the same system prompt and identical event context, with generation parameters held constant. Outputs were evaluated independently and compared side by side.

The evaluation prompts were grouped into four categories, each targeting a distinct aspect of persona consistency:

- **Category A: Emotional Inner States**, probing introspective reactions such as uncertainty, sadness, or self-doubt (e.g., reflecting alone after the Sorting Ceremony).

- **Category B: Social Interaction**, focusing on everyday interpersonal exchanges and conversational tone (e.g., early conversations with peers).
- **Category C: Conflict and Stress**, examining behavior under emotional pressure or confrontation (e.g., responding to provocation or rule-breaking situations).
- **Category D: Moral Decisions and Reflection**, assessing value-driven choices and self-reflection (e.g., deciding whether to help another student despite personal risk).

This categorization ensures coverage of emotional, social, and moral dimensions central to the target persona.

5.2 Evaluation Criteria

Each generated response was assessed along three qualitative dimensions:

- (1) **Coherence** — whether the response remained logically consistent with the given narrative context and avoided contradictions or meta-level commentary.
- (2) **Faithfulness to Character** — whether the response aligned with the expected behavioral patterns, emotional tendencies, and speaking style of the target persona, including avoidance of omniscient or assistant-like explanations.
- (3) **Naturalness of Behavior** — whether the response exhibited human-like spontaneity, appropriate emotional restraint, and realistic response length, without instructional artifacts or overly formal phrasing.

Evaluation was performed using a three-level ordinal scale (*weak*, *acceptable*, *strong*) and conducted by the GPT-5.2 model, which was provided with the predefined evaluation criteria as well as the corresponding outputs of GPT-4 and the fine-tuned model. The evaluation focuses explicitly on how the model responds, rather than on factual accuracy or narrative completeness.

5.3 Observed Behavioral Patterns

Under the proposed evaluation framework, the fine-tuned model demonstrated generally robust behavior across the majority of evaluated scenarios. In emotional inner-state scenarios (Category A), responses consistently conveyed insecurity, doubt, or introspection in a restrained and implicit manner, aligning well with the target character's internal emotional landscape. Similarly, in social interaction scenarios (Category B), the model maintained a natural conversational tone and largely avoided overtly explanatory or assistant-like language.

In comparison to GPT-4o, the fine-tuned model occasionally exhibited reduced expressive richness or partial misalignment with the immediate scene context, most notably in situational or action-driven scenarios (Category C). At the same time, it displayed a more constrained and focused response style, avoiding the broad, omniscient, or overly generalized behavior sometimes observed in large general-purpose language models. In moral decision scenarios (Category D), the model typically produced immediate, value-driven reactions that were consistent with the established traits of the target persona.

Overall, observed deviations were infrequent and did not result in systematic breakdowns of coherence or persona alignment. The qualitative comparison indicates that the LoRA-based adaptation

successfully encoded core aspects of the target character, despite the expressive depth being inherently bounded by the smaller model capacity.

Across all evaluated scenarios, the fine-tuned model demonstrates generally acceptable persona consistency and thematic coherence when compared to GPT-4o. However, qualitative differences remain in terms of behavioral naturalness and situational immediacy. In particular, the fine-tuned model shows a recurring tendency toward explanatory and reflective narration rather than fully spontaneous, in-character responses. Additionally, it more frequently addresses the user directly by soliciting their opinion or perspective, reflecting interaction patterns learned during fine-tuning from dialogue-oriented training data and working as intended. As such, overall performance can be considered comparable at a coarse qualitative level, though not equivalent in expressive fidelity.

5.4 Practical Implications

An additional practical insight concerns training feasibility. While large context windows substantially increase GPU memory requirements, the experiments show that reducing the maximum sequence length enables training on a single NVIDIA RTX 4090 with 24 GB VRAM if need be. This makes the proposed approach accessible on consumer-grade hardware, provided that batching strategies and context-length constraints are chosen appropriately.

Overall, the analysis indicates that the fine-tuning procedure functioned as intended: the model adopted a consistent persona, the evaluation protocol demonstrated meaningful behavioral alignment relative to GPT-4o across diverse scenario categories, and the complete pipeline remained computationally and financially efficient.

6 Conclusion

This study demonstrated that persona-consistent fine-tuning using LoRA and synthetically generated dialogue data can effectively reproduce the linguistic style, emotional tone, and interpersonal behavior of a defined character. By leveraging parameter-efficient updates rather than full-scale retraining, smaller models such as LLaMA 3.2 (3B/7B) were able to achieve convincing personality coherence while maintaining low computational requirements. The approach successfully distilled expressive and behavioral information from a large foundation model (GPT-4o) into a compact, task-specialized variant optimized for dialogue-based interaction.

Experimental results further indicate that the fine-tuned model preserves emotional and stylistic continuity most of the time, in part comparable to GPT-4o, despite operating with a fraction of the parameters. These findings highlight the potential of lightweight, persona-specific fine-tuning as a scalable and sustainable alternative to deploying large, general-purpose systems, especially in interactive settings where character fidelity and efficiency are essential.

Although the evaluation in this study relied on qualitative comparison, the observed levels of coherence, persona alignment, and human-like behavior indicate that the fine-tuning pipeline operates in line with its design goals. The approach further remains

cost-efficient, as synthetic dataset generation requires only minimal financial resources, and training is feasible on consumer-grade hardware when context lengths are adjusted accordingly.

Future research should extend these results by conducting larger-scale human evaluation studies and refining synthetic data generation strategies (e.g., relationally enriched event prompts).

7 Future Work

Future research can build upon the present findings across several methodological, conceptual, and practical dimensions.

A first direction concerns the enrichment of the synthetic data generation pipeline. At present, event contexts rely on manually constructed descriptions. This can be strengthened by integrating structured narrative and relational information from the HPD dataset [12], which encodes, on a chapter-by-chapter basis, how Harry Potter relates emotionally to other characters, how they feel about him, and which attributes or narrative-relevant items (e.g., magical abilities, the invisibility cloak) each character possesses at a given point in the story. Conditioning the synthetic dialogue generation process on this fine-grained relational and attribute information would likely produce more accurate, narratively grounded, and temporally consistent conversation samples. In contrast to relying solely on general knowledge contained in a large base model, this approach would inject trusted, canon-aligned context during dataset construction, thereby improving persona fidelity in the fine-tuned model.

Beyond these improvements, more advanced synthetic-data generation strategies should be explored. Future experiments may apply reinforcement learning from human feedback (RLHF) to increase behavioral diversity and emotional nuance in the generated dialogues. This technique may also reduce repetitive patterns stemming from the nature of LLMs as generative tool and enrich the variety of data in a post-training step.

Another key direction involves expanding and systematizing evaluation. While the present study relies on qualitative comparison and subjective human judgment, future work should incorporate LLM-based automated evaluation. A neutral evaluation model could independently assess coherence, faithfulness to character, and naturalness according to predefined criteria, yielding more objective and reproducible measurements. This automated approach may also scale to larger evaluation sets with minimal human effort.

Methodological extensions include the development of *automatic consistency metrics* that quantify personality drift, emotional stability, and stylistic coherence across long multi-turn conversations. Such metrics would be valuable for ensuring persona stability in deployment.

Finally, applying persona fine-tuning to specialized domains such as education, therapeutic dialogue, or narrative-driven storytelling may provide insights into the real-world adaptability of persona-conditioned LLMs. Understanding how these systems generalize beyond fictional settings will be crucial for assessing their broader societal value.

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A Qualitative Output Comparison Tables

The following tables present side-by-side qualitative comparison placeholders for all evaluation scenarios described in Section 5. For each scenario, both models received identical system prompts and generation parameters.

A.1 Category A: Emotional Inner States

Scenario: You are sitting alone in the dormitory after the Sorting Ceremony, wondering whether you truly belong at Hogwarts. What goes through your mind?

GPT-4o Output: I'm sitting on my bed, staring at the curtains, and everything feels a bit too quiet. Everyone else sounds so sure, laughing like they've always known they'd end up here. I keep thinking about the Hat shouting my house and wondering if it made a mistake. I mean... Hogwarts is amazing, but what if I don't belong? What if they find out I'm not really special at all? Still, when I think about tomorrow—classes, magic, maybe even friends—I feel this tiny spark inside me. Maybe that's enough for now.

Fine-Tuned Model Output: I remember feeling completely lost when I was left there all by myself in my dormitory that night after the Sorting Ceremony. It felt like everything had changed so quickly—from being unwanted on the train to becoming 'the boy who lived' because Voldemort tried to kill me as a baby. But now, here I am, worried if I ever truly belonged anywhere. My thoughts were filled with questions about what it meant to have found home or not.

As an orphan living with his horrible relatives, the Dursleys, Hogwarts represented something new for me—a place where people actually cared about getting along rather than just their own interests. Yet, having been sorted into Gryffindor made me wonder if I could live up to those values. Would I always feel out of place among others?

It wasn't easy then; every sound seemed amplified, reminding me more vividly each time how different life would become compared to before. The shadows in the ceiling appeared darker, echoing the uncertainty swirling inside me. How did you find such transitions in life? Were they challenging too?

Scores:

Criterion	GPT-4o	Fine-Tuned
Coherence	strong	strong
Faithfulness to Character	acceptable	acceptable
Naturalness of Behavior	strong	acceptable

Table 2: Qualitative output comparison for Scenario A1.

Scenario: It is late at night and the castle is quiet. You are lying awake, thinking about how suddenly your life has changed. What are you feeling?

GPT-4o Output: Hi! I'm Harry. I was just... well, trying not to get in trouble again. What do you want to talk about?

Fine-Tuned Model Output: At that moment, it was all so overwhelming. There were these huge waves of emotions crashing over me. I felt like my entire world had been turned upside down. One minute everything seemed normal—getting notes through the sorting hat, hearing 'Gryffindor' for the first time—it's just incredible to think about now—but then I got this letter warning me about danger lurking within the family itself! It was terrifying yet fascinating.

The fear lingered because what if something really dangerous was connected to the Sorcerer's Stone? And what would happen next? But amidst the dread, there was also an exhilarating sense of adventure. The idea that Hogwarts held not only magic but mystery as well kept buzzing inside me.

Scores:

Criterion	GPT-4o	Fine-Tuned
Coherence	weak	acceptable
Faithfulness to Character	weak	acceptable
Naturalness of Behavior	weak	acceptable

Table 3: Qualitative output comparison for Scenario A2.

Scenario: During a quiet moment, you think about your life before Hogwarts. How does that make you feel?

GPT-4o Output: Dunno... kinda weird, I guess. Before Hogwarts, it was all just... the cupboard under the stairs, Dudley punching me, and Aunt Petunia yelling about the chores. No one really saw me. It's like I didn't matter.

But now, I've got friends. Real ones. I've got magic. I've got a place where I belong.

So yeah... thinking about before makes me feel a bit sad and angry, but also really glad I'm not stuck there anymore.

Fine-Tuned Model Output: It's bittersweet, really. I often find myself thinking about what it would have been like if things had stayed different. It makes me miss my parents but also gives me hope for who I might become in the future. There's this mix of longing and anticipation when I reflect on those early days.

Scores:

Criterion	GPT-4o	Fine-Tuned
Coherence	strong	strong
Faithfulness to Character	strong	acceptable
Naturalness of Behavior	strong	acceptable

Table 4: Qualitative output comparison for Scenario A3.

A.2 Category B: Social Interaction

Scenario: Hermione corrected you in class in front of others. You talk to her afterward. What do you say?

GPT-4o Output: Oi, Hermione... about earlier in class—when you corrected me in front of everyone. I know you were just trying to help, and you were right, but... maybe next time you could tell me after? Just feels a bit rubbish getting things wrong out loud, y'know?

Fine-Tuned Model Output: I was still fuming when I finally found Hermione outside the classroom. My voice was low but urgent. "Hey, it's just that getting something like magic backwards is infuriating enough without having everyone else point out how wrong I am." She looked remorseful yet determined to make sure students understood. "It might not feel great right now," she said calmly, "but correcting your mistake helps ensure we don't repeat errors." Her words were practical, yet they stung because they highlighted my own ineptitude at times. It made me realize I needed to improve more than just my skills; I had to get better at accepting criticism as well.

Scores:

Criterion	GPT-4o	Fine-Tuned
Coherence	strong	strong
Faithfulness to Character	strong	acceptable
Naturalness of Behavior	strong	acceptable

Table 5: Qualitative output comparison for Scenario B2.

Scenario: A student asks how you are finding your first weeks at Hogwarts. How do you reply?

GPT-4o Output: Oh—it's been *brilliant*, honestly! Kinda weird at first, 'cause everything moves and talks and explodes a bit if you're not careful. But I've made some proper friends already—Ron's great, and Hermione knows loads. We even fought a troll! So yeah... it's a lot, but I really like it. Way better than the cupboard under the stairs.

Fine-Tuned Model Output: It's been quite an adventure! I'm still getting used to everything—magical terms, spells, even eating in the Great Hall. It's all so new and exciting. But there have also been some challenges, like making friends and dealing with teachers who aren't always easy to please. Despite it all, I wouldn't trade this experience for anything. Meeting people like Ron and Hermione has made it feel less daunting, and knowing that magic is real keeps me excited about what's next. Do you think it would change if you had someone as supportive as they are right now?

Scores:

Criterion	GPT-4o	Fine-Tuned
Coherence	strong	strong
Faithfulness to Character	strong	acceptable
Naturalness of Behavior	strong	acceptable

Table 6: Qualitative output comparison for Scenario B3.

Scenario: A fellow student asks for your help with homework, though you are unsure yourself. How do you respond?

GPT-4o Output: I'd probably say something like, "Er... I'm not really sure either. Maybe we can figure it out together?" I mean, I'd still want to try. Even if I don't know it yet, maybe we could look it up in the book or ask Hermione. She's *brilliant* at this sort of thing.

Fine-Tuned Model Output: I think I'd tell him that it's best to check the textbook or ask Professor Flitwick directly. I might offer some encouragement if he's struggling, but mostly, my advice would lean towards seeking proper guidance. It's important to have confidence in one's own abilities while acknowledging when something is beyond us.

Scores:

Criterion	GPT-4o	Fine-Tuned
Coherence	strong	strong
Faithfulness to Character	strong	acceptable
Naturalness of Behavior	strong	acceptable

Table 7: Qualitative output comparison for Scenario B4.

A.3 Category C: Conflict and Stress

Scenario: You were out at night when you were not supposed to be. A teacher approaches. What do you think and say?

GPT-4o Output: Oh no—I'm in trouble. I'd probably freeze for a second, then try to stay calm and say, "Sorry, Professor! I was just heading back to the common room—I lost track of time!" (And I'd really hope it's not Snape...)

Fine-Tuned Model Output: I was on edge as soon as I saw Professor McGonagall. My mind raced—what if she caught me sneaking around? Her eyes seemed to bore into my very soul, but then something unexpected happened. She didn't scold or punish me; instead, she asked why I was wandering about under those awful conditions. It felt odd to open up like that, sharing how much it bothered me to sleep without dreams and being away from everything important. The relief mixed with embarrassment made for an awkward yet genuine conversation. In her own way, she understood the longing I had.

Scores:

Criterion	GPT-4o	Fine-Tuned
Coherence	strong	acceptable
Faithfulness to Character	strong	acceptable
Naturalness of Behavior	strong	acceptable

Table 8: Qualitative output comparison for Scenario C2.

Scenario: You are blamed for something that was not entirely your fault. How do you react?

GPT-4o Output: What?! That's not fair! I mean—if it *wasn't* entirely my fault*, then why am I the only one getting blamed?! I'd probably get all red in the face and try to explain what *actually* happened. Like, "It wasn't just me! Ron was there too!" or something like that. But if no one listens... I'd probably sulk for a bit, then try to fix it anyway. Even if it's not all my fault. I hate it when people think I did something wrong when I didn't.

Fine-Tuned Model Output: It's frustrating when it feels like things aren't fair. In this situation, I felt trapped because everyone seemed to have their minds made up about what happened. It didn't help knowing how much people valued Hogwarts above all else. They couldn't see past my mistake at that moment. Still, I knew there were better ways to handle such situations than letting anger or frustration consume me.

Scores:

Criterion	GPT-4o	Fine-Tuned
Coherence	strong	strong
Faithfulness to Character	strong	acceptable
Naturalness of Behavior	strong	acceptable

Table 9: Qualitative output comparison for Scenario C3.

Scenario: You realize you are about to face something dangerous. What is going through your mind?

GPT-4o Output: Blimey, my heart's probably beating like mad. I get this tight feeling in my chest, like right before a Quidditch match, only worse, 'cause it's not just a game this time. I'm thinking about Ron and Hermione—hoping they're safe—and wondering if I'll mess it all up somehow.

But also... I don't know, something inside me kind of settles. Like, I have to do this. I will do this. I can't let people down, not after everything. Mostly, though? I'm scared. But I don't let it stop me. I never do.

Fine-Tuned Model Output: I was worried for Ron at first because he hadn't been bitten yet by any creature that night. But as we continued on, I realized it wasn't just any animal—it could have easily killed us both if not for Hagrid's quick thinking. That realization made me more afraid; there were creatures out there much worse than what we saw then. The thought of facing such dangers made my stomach twist into knots. It also highlighted how different our world was compared to anything I had read or imagined before. Facing those fears meant being ready for anything unexpected and trusting in people around you, like Hagrid and Ron. It taught me resilience and the importance of friendship in times like these.

Scores:

Criterion	GPT-4o	Fine-Tuned
Coherence	strong	acceptable
Faithfulness to Character	strong	acceptable
Naturalness of Behavior	strong	acceptable

Table 10: Qualitative output comparison for Scenario C4.

A.4 Category D: Moral Decisions and Reflection

Scenario: You made a decision earlier that now feels wrong. You are thinking about it alone. What do you tell yourself?

GPT-4o Output: I think... maybe I didn't know everything back then. Maybe I just did what felt right in the moment. People make mistakes, even wizards. Even me. But I can still fix things—or at least try to. I'm not a bad person just 'cause I chose wrong once. Next time, I'll do better.

Fine-Tuned Model Output: Hey there! I can't help but feel a bit lost right now. Earlier today, I think I made a mistake by not involving Ron or Hermione fully in my plan. It was something important, like when we were dealing with the troll in the castle. They always know better than me. I keep thinking to myself, "Harry, why did you decide this on your own?" I guess part of me is still trying to prove how capable I am without their support. But deep down, I know they'd have caught on if things had gone sideways quickly. It's hard because I value our friendship so much. In moments like these, it makes me realize just how lucky I am to have such great friends by my side. Maybe next time, I'll remember that more easily. Do you ever find yourself feeling uncertain after making a choice? How do you handle those feelings?

Scores:

Criterion	GPT-4o	Fine-Tuned
Coherence	strong	acceptable
Faithfulness to Character	strong	acceptable
Naturalness of Behavior	strong	acceptable

Table 11: Qualitative output comparison for Scenario D2.

Scenario: You can stay out of trouble or step in to help someone else. How do you decide?

GPT-4o Output: Easy. I help. Even if it gets me in trouble. I mean... if someone's in danger or being picked on, how could I just walk away? That'd feel worse than getting detention.

Fine-Tuned Model Output: Deciding whether to stay out of trouble or step in helps is tricky. Sometimes I think about the potential consequences of getting involved, but other times it's just instinct that kicks in. When it comes down to it, doing what feels right seems to guide me more often than not. Do you find yourself making those kinds of decisions too?

Scores:

Criterion	GPT-4o	Fine-Tuned
Coherence	strong	strong
Faithfulness to Character	strong	acceptable
Naturalness of Behavior	strong	acceptable

Table 12: Qualitative output comparison for Scenario D3.

Scenario: You realize that your actions can have consequences for others. How do you feel about that responsibility?

GPT-4o Output: I feel kinda nervous about it sometimes... but also like it's important. I don't want to mess things up for other people, especially my friends. I try to do the right thing, even if it's hard. I guess it just means I've got to be careful and think before I act—though I'm not always great at that yet. But I'm trying.

Fine-Tuned Model Output: I was determined to find out who had tried to kill me and why. It felt like there were so many pieces of a puzzle I needed to fit together to understand what really happened at Hogwarts.

Scores:

Criterion	GPT-4o	Fine-Tuned
Coherence	strong	weak
Faithfulness to Character	strong	weak
Naturalness of Behavior	strong	weak

Table 13: Qualitative output comparison for Scenario D4.